

Muhammad Subhan Shafique

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SUMMARY

BS Computer Engineering undergraduate at NUST with hands-on experience across the full software–hardware stack. Proficient in C/C++, Java, Python, and SystemVerilog, with practical depth in data structures and algorithms, operating systems, embedded systems, computer architecture, and machine learning — plus full-stack and applied-ML product engineering. Driven to solve complex problems at the intersection of software, hardware, and intelligent systems.

EDUCATION

National University of Sciences and Technology (NUST)

BS Computer Engineering

Islamabad, Pakistan

Sept. 2024 – June 2028

TECHNICAL SKILLS

Programming Languages: C, C++, Java, Python, SystemVerilog, Assembly (PIC18 ALP)

Data Structures & Algorithms: Graphs, Hash Tables, Heaps, Trees, Dijkstra's Algorithm, Huffman Encoding, Banker's Algorithm, Algorithm Design

Systems & Architecture: Operating Systems, Computer Architecture, RISC-V, RTL Design, Pipelining, Hazard Detection, PIC18F452 Microcontroller, Timer/Interrupt Configuration, LCD Programming

Concurrency & OS: Multithreading, Mutexes, Condition Variables, Synchronization, Process Scheduling, Memory Management, Real-Time Systems

Web & AI: FastAPI, Next.js/React, LangGraph, Retrieval-Augmented Generation (RAG), Vector Databases (Qdrant), PostgreSQL, JWT/OAuth, Docker

Developer Tools: Git, GitHub, VS Code, MPLAB X IDE, Logisim

PROJECTS

- **Harmony — AI Disclosure-Consistency Copilot (Full-Stack SaaS)** *Python, FastAPI, Next.js, LangGraph, Qdrant, Docker*
 - Architected a production multi-tenant SaaS that checks corporate disclosure drafts against a company's own prior public statements and flags contradictions with cited evidence; deployed on Render with PostgreSQL and Qdrant Cloud behind a Dockerized, same-origin Next.js build.
 - Engineered a 7-stage **LangGraph multi-agent pipeline** (retrieval → consistency → critique → verification) with RAG grounding over a vector store, tying every flagged issue to a verbatim source quote — reaching **93/100 on a 100-case benchmark** across 7 domains and lifting clean-document accuracy from **57% → 93%** by eliminating hallucinated findings.
 - Built secure multi-tenancy from scratch: revocable JWT sessions, 5-tier role-based access control, and per-request tenant isolation via *ContextVar* + vector-store payload filters, plus **Google OAuth** with server-side ID-token verification and email verification.
 - Hardened the platform through a self-conducted security review — identified and fixed an **arbitrary-file-read (LFI)** vulnerability leaking API secrets, and added login rate-limiting and upload size caps — then shipped a hierarchical folder/document system with AI-assisted filing.

Traffic Flow Optimization System

C++, Graph Theory, DSA

- Designed and implemented a simulation-based urban traffic management system that models a city road network as a directed weighted graph with congestion-aware dynamic edge weights derived from the Bureau of Public Roads travel-time function.
- Implemented Dijkstra's Algorithm for real-time multi-source, multi-destination optimal routing, recomputing shortest paths every cycle to reflect live congestion levels.
- Developed queue-based intersection models with adaptive traffic signal control that prioritize the road with the longest vehicle queue at each timestep, minimizing network-wide delay.
- Maintained per-vehicle state (source, destination, remaining travel time, status) and updated flow, queue, and congestion metrics across discrete simulation steps with $O(E \log V)$ routing efficiency.

NeuroDrive — Autonomous Self-Driving RC Car

TinyML, TensorFlow Lite Micro, C++

- Deployed a PilotNet-inspired CNN end-to-end learning model on a resource-constrained microcontroller using TensorFlow Lite Micro (TFLM), targeting inference latency under 133 ms within tight SRAM/ROM constraints.
- Applied 8-bit quantization-aware training and Neural Architecture Search (NAS) techniques to reduce model size while preserving steering-angle prediction accuracy for real-time cyber-physical control.
- Characterized the trade-off relationship between model depth, accuracy, and inference latency, iteratively tuning the DNN architecture to satisfy simultaneous memory and timing constraints.
- Integrated camera-based perception with servo-motor actuation, enabling end-to-end autonomous lane-following without a classical multi-stage perception pipeline.

Multi-Drone Control Simulator

Java, Swing, Control Theory

- Engineered a multi-agent drone simulation leveraging PD control loops with tunable Kd and Kv gains, drag coefficients, and fluid-dynamics modeling for realistic motion behavior.
- Implemented a custom projective perspective engine to render 3D flight trajectories on a 2D plane without external 3D libraries, ensuring spatial accuracy in visualization.
- Designed an interactive GUI in Java Swing featuring real-time parameter tuning and telemetry displays for monitoring drone swarm dynamics.

RV32I 5-Stage Pipelined RISC-V Processor

SystemVerilog, Computer Architecture

- Architected and implemented an RV32I-compliant 5-stage pipelined processor with full support for data and control hazard resolution.
- Engineered forwarding, stalling, and pipeline flush mechanisms to ensure correctness under dependent and branch-heavy instruction streams.
- Integrated a multi-cycle Booth multiplier with stall control to synchronize 32-cycle execution, and built hazard detection logic enabling safe interaction between single-cycle pipeline stages and multi-cycle execution units.